

diode D1 and the parallel combination of resistor R9 and preset VR3. The base of pnp transistor T5 connects directly to collector of T3. The driver stage therefore drives a relatively low-resistance load, requiring a transistor capable of handling high power.

The output transistors are npn transistor T4 and pnp transistor T5, connected as a complementary symmetry class-AB output stage. In this configuration, an npn transistor and a pnp transistor (complementary) with equal current gains (symmetrical) are required. Thus, ideally, the DC current gains of T4 and T5 should be matched by measurement.

The DC biasing circuit for T4 and T5 has one diode and two parallel-connected resistors. Preset VR3 is used to adjust the quiescent collector current of T4 and T5 and therefore the class of operation. The output stage also involves capacitor C4, which is known as a bootstrapping capacitor. Bootstrapping is included to allow a higher output voltage swing. If capacitor C4 is not included, biasing resistors R7 and R8 are combined into a single resistor.

Ideally, the output signal should be able to swing from 0V to the value of the supply voltage. However, this cannot happen due to the 0.6V forward

bias required across the base-emitter junctions of the output transistors and also the losses. For the positive half cycle, if the output is to reach the supply voltage, the voltage at T4 must be at least 0.6V higher than the supply voltage. Similarly, an output of 0V can be obtained only if the base voltage of T5 falls to -0.6V.

By adding bootstrapping capacitor C4, the output voltage swing is effectively added to the DC bias voltages. Thus, for the positive half-cycle, the positive change adds to the bias voltage at T4, causing it to conduct more current and produce a higher output voltage. Similarly, in the negative half-cycle, the negative-going swing reduces the quiescent bias voltage, helping T5 turn on harder and produce a lower output voltage.

An important aspect of amplifier design is power supply decoupling. When the output stage is producing full output power, the power supply delivers substantial peak currents. Under these conditions, it is possible that some of the audio signal appears on the supply line. To prevent this signal from affecting the operation of the rest of the circuit, it must be eliminated from that part of the supply feeding the voltage amplifier section. So resistor R6 is added to-

gether with C2 and ZD1 to maintain the amplifier's supply voltage to a constant 10V.

Assemble the circuit on a general-purpose PCB and enclose in a suitable cabinet. Mount the diodes, electrolytic capacitors and transistors with the correct polarity. Output transistors T4 and T5 have a metalised surface on one side, which should face the centre of the board. A heat-sink is required for both the transistors. Either a small (20mm²) piece of aluminium or a commercially available heat-sink can be used. The heat-sinks on T4 and T5 should be insulated from the transistors with a piece of Mylar or similar material, as this transistor (and therefore the heat-sink) connects directly to the power supply.

Before applying power to the circuit, connect an 8-ohm load (resistor or loudspeaker) to the output, and capacitor C1 between the input terminal and the volume control. Set the volume control to minimum and then apply power—either from the plug pack or an external 12V DC supply.

Ensure that both the output transistors are cool when touched. If not, try adjusting preset VR3. The correct setting for VR3 should give a quiescent collector current of around 100 mA through T4 and T5. ●

Simple Stereo Level Indicator

PETRE TZV. PETROV

Usually, low-priced home stereo power amplifiers don't have output level indicators. An output power level indicator can be added to each channel of these stereo power amplifiers. As low levels of the output power are not disturbing and damaging to the people, there is no need to add a preamplifier and low-level detector before IC LM3915. But you should know when the output power becomes considerably high.

Here we present a very simple, low-cost stereo-level indicator circuit for home power amplifiers with power rating of around 0.5W. The circuit is built around two LM3915 dot/bar display driver ICs (IC1 and IC2). LM3915 senses analogue voltage levels to drive ten LEDs, providing a logarithmic 3dB/step analogue display.

The voltage levels below 1V are not important because these correspond to a low level of the audio signal. Similarly, input voltage levels above

30V correspond to too high levels of the output power, which are not applicable for home power amplifiers. So the voltage levels of our interest are 1V to 30V, which can be handled directly with LM3915. LM3915 needs no protection against $\pm 35V$ inputs, which simplifies the circuit.

Most audio power amplifiers can drive 2-ohm to around 32-ohm loads. A load of several kilo-ohms will not change the conditions for the ampli-

